



ECE 501: Digital Image Processing

Week 3 Progress Report

Group 6

Vehicle Orientation Estimation Using Conventional Image Processing

Submitted to faculty: Prof. Mehul Raval

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GitHub Repository (DRASHTI-HaOBB):

<https://github.com/ERYAGNIK003/DRASHTI-Dataset>

Dataset Source:

DRASHTI-HaOBB – Heading-angle Oriented Bounding Box Dataset

<https://zenodo.org/records/18278989>

1. Work Done This Week

In Week 3, we moved from vehicle crop extraction to the actual orientation-estimation stage of the project. Based on the work completed in Week 2, the same vehicle crop extraction process was continued using the DRASHTI-HaOBB OBB annotations.

The main objective this week was to compare three conventional image-processing methods for estimating vehicle orientation:

1. PCA-based orientation estimation
2. Hough Transform-based orientation estimation
3. Contour-based orientation estimation

First, the complete process was tested on one vehicle image. The vehicle was extracted using the OBB coordinates, and its original orientation was preserved. The ground-truth heading was converted from the 0° – 359° range into the 0° – 90° orientation range required for the project.

A common preprocessing pipeline was then applied before the three methods. The preprocessing included grayscale conversion, CLAHE-based contrast enhancement, Gaussian smoothing, Canny edge detection and morphological closing. The same preprocessing was used for all three methods so that the comparison was based mainly on the difference between the orientation-estimation methods.

For the PCA-based method, the contour points obtained from the processed image were used to find the principal direction of the vehicle shape. The principal axis obtained from PCA was then converted into the 0° – 90° orientation range.

For the Hough Transform method, line segments were detected from the edge image using the probabilistic Hough Transform. The detected line orientations were combined using line length as a weight to obtain a dominant vehicle orientation.

For the contour-based method, the selected vehicle contour was fitted with a minimum-area rotated rectangle. The orientation of the long axis of this rectangle was used as the estimated vehicle orientation.

After the single-image test, exactly 100 different training images were selected. One valid vehicle with difficulty value 0 was selected from each image. The same 100 vehicle samples were then processed using all three methods. For each vehicle, the ground-truth orientation, estimated orientation from each method and angular error were recorded in a CSV file.

The complete Week-3 pipeline was therefore:

Dataset → Annotation Parsing → Image/Annotation Matching → OBB-based Vehicle Crop Extraction → Common Preprocessing → PCA / Hough Transform / Contour-based Estimation → Angular Error Calculation.

2. Methodology

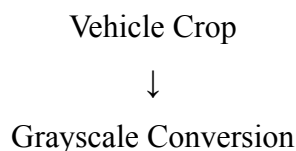
The methodology used in Week 3 is divided into the following stages.

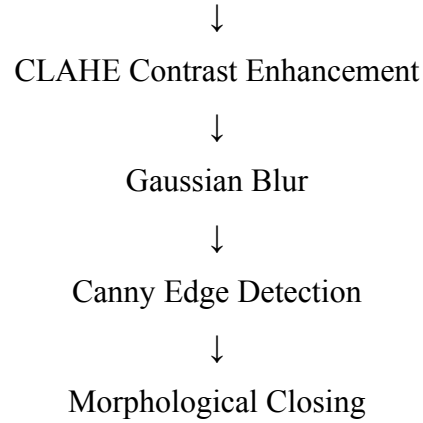
2.1 Vehicle Crop Extraction

The OBB annotation was used to locate the vehicle in the original training image. An axis-aligned crop was then created around the OBB with a small amount of padding. The vehicle was not rotated during cropping so that its original orientation was preserved for estimation.

2.2 Common Image Preprocessing

The same preprocessing pipeline was applied to all three methods.





The resulting edge image was used for extracting shape and line information.

2.3 PCA-based Orientation Estimation

PCA was applied to the selected vehicle contour points to determine the principal direction of the vehicle shape. The first principal component was treated as the dominant axis of the vehicle. The obtained angle was then converted into the 0° – 90° orientation representation.

2.4 Hough Transform-based Orientation Estimation

The probabilistic Hough Transform was applied to the edge image to detect line segments. The angle of each detected line was calculated, and line lengths were used as weights while combining the line orientations. The resulting dominant orientation was converted into the 0° – 90° range.

2.5 Contour-based Orientation Estimation

The selected vehicle contour was fitted using a minimum-area rotated rectangle. The long-axis direction of the rectangle was taken as the estimated vehicle orientation.

2.6 Angular Error

The predicted orientation from each method was compared with the ground-truth orientation. Since both values were represented in the 0° – 90° range, the angular error used in this experiment was calculated as the absolute difference between the estimated orientation and the ground-truth orientation.

$$Angular\ Error = | \theta_{predicted} - \theta_{ground\ truth} |$$

where,

$\theta_{predicted}$ = orientation estimated by PCA, Hough, or Contour

$\theta_{ground\ truth}$ = ground-truth orientation after converting the original heading to the required $0^{\circ} - 90^{\circ}$ range

3. Results and Observations

The three methods were tested on the same set of 100 vehicle images. All selected samples had difficulty value 0.

The result CSV contained 100 vehicle samples. The sample included 11 vehicle classes. The distribution was not equal across all classes. SUV was the most frequent class with 34 samples, followed by MotorCycle with 30 samples. AutoRicksaw had 8 samples, Sedan had 7, Truck had 5 and PickUp had 4. Auto3WCargo, Container and Trailer each had 3 samples, Bus had 2 samples and Tanker had 1 sample. The classes Mixer, Tipper and Van were not present in this particular 100-image sample.

3.1 Overall Method Comparison

The results obtained from the CSV are shown below.

Method	Successful Predictions	Failed Predictions	Success Rate	Mean Angular Error	Median Angular Error	Standard Deviation	Maximum Error
PCA	100	0	100%	10.74°	4.85°	16.11°	81.71°
Hough Transform	98	2	98%	12.46°	4.18°	20.44°	82.39°
Contour-based	100	0	100%	10.97°	5.72°	16.44°	89.00°

(Table 1. Overall comparison of the three orientation-estimation methods)

The PCA method produced a mean angular error of 10.74° over the 100 samples, with all 100 images producing a valid prediction. The median error was 4.85° , which shows that the typical error was smaller than the mean because some images had much larger errors.

The contour-based method produced a mean angular error of 10.97° and also produced valid predictions for all 100 images. Its median error was 5.72° .

The Hough Transform method produced a mean angular error of 12.46° . It successfully produced predictions for 98 of the 100 images, while two images did not produce a valid Hough orientation. Its median error was 4.18° , which was slightly lower than the median errors of PCA and the contour-based method. However, its mean error and standard deviation were higher, showing that some larger-error cases affected its overall performance.

From the 100-image experiment, PCA and contour-based estimation showed similar overall mean errors, while the Hough-based method showed a higher mean error and two failed predictions in this sample.

3.2 Error Distribution

The CSV results also show that:

Error Limit	PCA	Hough Transform	Contour-based
Error $\leq 5^\circ$	50 / 100	55 / 98	47 / 100
Error $\leq 10^\circ$	74 / 100	71 / 98	76 / 100
Error $\leq 20^\circ$	86 / 100	86 / 98	84 / 100

In this 100-image sample, the three methods produced relatively small errors for a substantial portion of the valid predictions, while a smaller number of samples produced substantially larger errors.

3.3 Class-wise Observations

The class-wise results show that the performance was different for different vehicle types.

For the classes with more samples, PCA showed mean errors of 7.28° for MotorCycle, 7.64° for SUV and 7.49° for AutoRicksaw. The contour-based method showed mean errors of 10.15° , 8.34° and 8.29° for the same classes, respectively. The Hough method showed mean errors of 4.54° for MotorCycle, 12.97° for SUV and 13.97° for AutoRicksaw.

The Sedan class had a mean error of 23.55° with PCA, 7.03° with Hough and 18.10° with the contour-based method.

For Container samples, the mean errors were 40.12° with PCA, 49.13° with Hough on the two valid predictions, and 27.49° with the contour-based method.

For Bus, the Hough method showed a mean error of 0.74° in the two available samples, while PCA and contour-based estimation showed mean errors of 27.32° and 24.18° , respectively.

The results also show that some classes had very few samples. For example, Tanker had only one sample, while Bus had two samples. Therefore, these class-wise values should be treated as initial observations from the 100-image experiment and not as final conclusions about the full dataset.

3.4 Single-image Testing

Before the 100-image experiment, the three methods were tested on one vehicle crop.

The single-image test confirmed that:

- The vehicle could be correctly extracted using the OBB.
- The original vehicle orientation was preserved.
- The same preprocessing pipeline could be applied.
- PCA could produce an orientation estimate.
- Hough Transform could produce an orientation estimate when sufficient lines were detected.
- The contour-based method could produce an orientation estimate from the vehicle contour.

This single-image validation was completed before moving to the larger 100-image experiment.

3.5 Main Observations from Week 3

Based on the 100-image CSV results, the main observations are:

1. All three conventional methods were successfully implemented and tested on the same vehicle samples.
2. PCA produced a mean angular error of 10.74° and successfully processed all 100 samples.
3. The contour-based method produced a similar mean angular error of 10.97° and also successfully processed all 100 samples.
4. Hough Transform produced a higher mean angular error of 12.46° and failed to produce a valid estimate for 2 of the 100 samples.
5. The median errors of all three methods were relatively low compared with their maximum errors, showing that some individual images had much larger errors than the typical sample.
6. The three methods behaved differently for different vehicle classes. This suggests that vehicle shape and visible edges may affect the orientation-estimation performance.
7. The current 100-image sample is not equally distributed among all vehicle classes, so the class-wise results are useful for initial observation but will need a larger and more balanced sample in later stages.
8. Large errors are present in some images, showing that the current basic methods can still be affected by vehicle shape, background edges, weak contours or other image conditions.

4. Challenges Faced

The main challenge this week was implementing the three methods so that they could operate on the same vehicle crops and use the same preprocessing pipeline.

Another challenge was handling the different output behaviour of the methods. PCA and contour-based estimation produced predictions for all 100 selected vehicles, whereas the Hough Transform did not produce a valid prediction for two vehicles.

The methods also produced some large errors. This shows that edges and contours from the surrounding image can influence the orientation estimate, especially when the vehicle shape is not clearly separated from its surroundings.

Another challenge was that the 100-image sample was not equally distributed across the vehicle classes. Some classes had many samples while some classes had very few or no samples in the selected set. This needs to be considered in later analysis.

5. Current Status

The first baseline orientation-estimation stage has now been completed.

We have:

- Tested the complete process on one vehicle image.
- Applied common preprocessing to the vehicle crop.
- Implemented PCA-based orientation estimation.
- Implemented Hough Transform-based orientation estimation.
- Implemented contour-based orientation estimation.
- Selected exactly 100 different training images.
- Extracted one valid vehicle from each selected image.
- Applied all three methods to the same 100 vehicle samples.
- Compared the estimated orientation with the ground-truth orientation.
- Calculated angular error for each method.
- Created a combined CSV containing the 100-image results.

The current results show that PCA and contour-based methods produced similar mean angular errors, while Hough Transform had a higher mean error and two failed predictions in the current sample.

The orientation-estimation stage will be improved further in the next stage using the results and error cases obtained from this experiment.

6. Work to be done next week

In the next stage, we plan to:

- Study the images with large orientation errors.
- Identify the reasons for incorrect orientation estimates.
- Improve the preprocessing and contour/edge extraction where required.
- Investigate the parameters used in PCA, Hough Transform and contour-based methods.
- Test the methods on a larger number of vehicle samples.
- Continue comparing the three methods using the obtained error results.
- Start a more detailed analysis based on vehicle class, vehicle size, image quality and orientation.

7. References

- [1] Y. Bhavsar, M. Zaveri, M. Raval, K. R. Patel, and S. B. Zaveri, “Descriptor: Drone Nadir-view Annotated Images of Vehicles Detection Dataset for India with Heading-angle Oriented Bounding Box (DRASHTI-HaOBB),” IEEE Data Descriptions, 2026, doi:<https://ieeexplore.ieee.org/document/11421776>.
- [2] Y. Bhavsar, M. Zaveri, M. Raval, S. Zaveri, and Ahmedabad University, “DRASHTI-HaOBB: Drone nadir-view Annotated images of vehicles dataset for India - Heading-angle Oriented Bounding Box,” Zenodo, 2026, doi:<https://zenodo.org/records/18278989>.